

2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used:

Type	Method Applied	Features Processed
Categorical Variables	Label Encoding (LabelEncoder)	Attrition (<i>Target</i>), BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime

- **Implementation Detail:** All object-type categorical columns identified in the dataset were mapped to numerical representations using LabelEncoder().

2.2 Feature Transformation

- **Outlier Removal:** Applied Interquartile Range (IQR) filtering across all numerical columns (num). Rows containing values falling outside the lower boundary $Q1 - 1.5 * IQR$ or upper boundary $Q3 + 1.5 * IQR$ were filtered out.
- **Feature Scaling:** Applied **Standardization** (StandardScaler) to all feature inputs X). This transformed features to have zero mean and unit variance , ensuring optimal convergence for distance and gradient-based algorithms.

2.3 Feature Selection

Removed:

- **Non-Informative / Constant Features:**
 - EmployeeCount (*Constant value = 1*)
 - StandardHours (*Constant value = 80*)
 - Over18 (*Constant value = 'Y'*)
 - EmployeeNumber (*Unique identification index*)

2.4 Final Feature Set

- **Dataset Split:** Dataset split into **80% Training Data** (x_train, y_train) and **20% Testing Data** (x_test, y_test).

- **Input Predictor Matrix (X):** 30 cleaned, filtered, encoded, and standardized predictor features:
 - Age, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, EnvironmentSatisfaction, Gender, HourlyRate, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MaritalStatus, MonthlyIncome, MonthlyRate, NumCompaniesWorked, OverTime, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager
- **Target Variable (y):**
 - Attrition (Label Encoded: 1 -> Yes, 0 -> No)